

“NEWS AGGREGATOR SYSTEM”

MINI PROJECT 1B

BACHELOR OF COMPUTER SCIENCE AND DESIGN

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Mini-Project -1B Report Approval for S.E.

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2024-25.

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Abstract

In the digital age, the rapid growth of online news creates an overwhelming amount of information, making it difficult for users to stay informed. This project focuses on automatic news summarization using Natural Language Processing (NLP) with an abstractive approach to address this challenge. Unlike extractive methods that pull key sentences directly from the text, abstractive summarization generates new sentences by understanding and rephrasing the content.

The system uses advanced transformer-based models like BART and T5 to process news articles, extract the core message, and produce concise, human-like summaries. The workflow involves data collection from news sources, text preprocessing, model-based summarization, and post-processing for coherence and fluency. The model's performance is evaluated using metrics like ROUGE and BLEU to ensure accuracy and relevance.

This project has practical applications in news aggregation platforms, media monitoring, and content curation, enabling users to access essential information quickly. Future enhancements include multi-document summarization, improved coherence, and personalized summary generation for better user experience.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's fast-paced world, staying informed about the latest news and events is more important than ever. However, with the overwhelming amount of information available, it can be challenging to keep up with everything. This is where news aggregation comes in – the process of creating a condensed version of a news article or document that captures the essential information and key points.

With the rise of natural language processing (NLP) techniques and tools, automatic news summarization has become a popular area of research. The ability to extract the most important information from a large News corpus and present it in a concise format has many potential applications, from news aggregation and personalized content recommendation to legal document analysis and financial news monitoring.

News Aggregator system is the process of condensing a large news article or document into a shorter version, while retaining the important information and conveying the main idea. With the advent of NLP techniques, automatic news summarization has become a popular area of research. In this min project, the various NLP techniques and tools for news summarization are used. We will start by collecting a corpus of news articles, and pre-process the data to clean and to tokenize the News. The challenges and limitations of news summarization, such as dealing with ambiguity, maintaining coherence, and preserving the author's intent can be identified and managed.

CHAPTER 02

LITERATURE SURVEY

2.1 News Aggregator System

A **News Aggregator System** collects and summarizes news articles from multiple sources, providing users with a consolidated and organized view of current events. This system uses **Natural Language Processing (NLP)** and **Machine Learning (ML)** techniques to extract, summarize, and present key information efficiently. The following literature survey reviews significant research and technological advancements in the development of automated news aggregation and summarization systems:

1. Traditional News Aggregation Approaches
2. Modern NLP Techniques for News Summarization
3. News Aggregation Using APIs
4. Evaluation Metrics for News Summarization
5. Future Directions in News Aggregation Systems

The evolution of **news aggregator systems** has advanced from **rule-based** techniques to **deep learning** models capable of generating **abstractive** summaries. While modern methods like **BART** and **T5** offer high-quality outputs, challenges remain in ensuring factual accuracy, reducing computational costs, and delivering personalized content. Future improvements will focus on **multi-source integration**, **better evaluation methods**, and **user-centric delivery**.

Limitations

1. **Data Quality** – Risk of misinformation and inconsistency across sources.
2. **Summarization** – Loss of context, accuracy issues, and information overlap.
3. **Technical** – High computational demands, scalability issues, and latency.
4. **Personalization** – Bias reinforcement and limited content diversity.
5. **Evaluation** – Challenges in accurately assessing and monitoring content.

While **news aggregator systems** offer efficient and automated ways to consolidate news from multiple sources, they still face several challenges and limitations.

2.2 Objectives

The primary goal of a News Aggregator System is to collect, process, and present concise and accurate summaries of news articles from multiple sources. This system enhances information accessibility, saves user time, and ensures diverse perspectives. The following are the key objectives:

1. **Automate** the collection of news articles from diverse sources.
2. **Summarize** news accurately using advanced **NLP** techniques.
3. **Enhance user experience** with organized and personalized content.
4. **Ensure diversity** and **accuracy** in the summarized output.
5. **Evaluate and optimize** system performance with robust metrics.
6. **Deliver flexible summaries** to cater to various user needs

2.3 Future Scope of the News Aggregator System

As technology advances and user demands evolve, there are numerous opportunities to enhance the News Aggregator System. Future improvements can focus on increasing accuracy, enhancing user experience, and integrating advanced AI capabilities. Below are key areas for future development:

1. User Authentication & Personalization:
2. News Recommendations:
3. Multi-Language Support:
4. Voice Search and Text-to-Speech:
5. Live News Updates & Streaming:
6. Offline Access:
7. Collaborations with Journalists & Publications:
8. Advanced Search & Filters:
9. Mobile App Integration:
10. Monetization Opportunities:.

CHAPTER 3

PROPOSED SYSTEM

3.1 Proposed System

The proposed system consists of several key components to ensure efficiency, scalability, and user engagement.

4.1 Environment: The system is deployed on a web server, allowing users to access the platform via a browser on desktop and mobile devices. The backend is built with Django, ensuring smooth communication with the GNews API, while the front end is designed to be lightweight and responsive.

4.2 User Interface: A well-structured and intuitive interface allows users to browse, search, and filter news articles effortlessly. The UI is designed using HTML, CSS, and JavaScript, ensuring an optimal user experience.

4.3 Real-time News Updates: The system fetches news from the GNews API at regular intervals to ensure users have access to the latest articles from various sources.

3.4 News Categorization: Articles are categorized by topics such as politics, technology, business, sports, and entertainment. This allows users to quickly find content relevant to their interests.

3.5 Search and Filtering Features: Users can search for specific news articles using keywords or filter articles by date, source, or category, improving accessibility and user experience.

3.6 Responsive Design: The system is fully responsive, allowing users to access the platform on desktops, tablets, and smartphones without any performance issues.

3.7 Security Measures: Basic security measures, such as API request validation, error handling, and secure content rendering, are implemented to ensure data integrity and protection against unauthorized access.

3.8 Future Enhancements: The system is designed to be scalable, with potential future enhancements such as AI-driven content recommendations, user authentication for personalized feeds, and multilingual support.

WORKFLOW



Fig 1: Flowchart

CHAPTER 4

IMPLEMENTATION

4.1 Hardware Software Requirment

1. Hardware

Windows Requirements

RAM: 4 GB

CPU: Intel Pentium

Web Browser: chrome, firefox, etc.

OS: Windows 8

Android Requirements

RAM: 4 GB

processor: Snapdragon 425

Network: Internet Connection 4g/5g.

OS: Android 10

2. Software

1. Integrated Development Environment (IDE): Visual Studio Code (VS Code) Version: 1.83.1.

2. Backend: i) Python 3.13.2 ii) Django framework 5.1.7

3. API Integration: GNews API (for fetching real-time news data)

4. Markup Language: Hypertext Markup Language (HTML5) Version: 4.01.

5. Styling: Cascading Style Sheets (CSS) Version W3.CSS.

6. Scripting language: JavaScript (JS) Version: ES2024.

4.2 Implementation Plan

Week 1	Discussion of topic
Week 2	Gathering information & learning language
Week 3	Confirmation of topic
Week 4	Project Setup
Week 5	API Integration
Week 6	Core Features Development
Week 7	User Interface
Week 8	System Optimization

CHAPTER 6

Expected Outcome

The Daily Breeze

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CONCLUSION

The **News Aggregator System** provides an efficient and automated platform to collect, summarize, and present news from multiple sources. By leveraging **Natural Language Processing (NLP)** and **machine learning** techniques, the system delivers concise and accurate summaries, enhancing user accessibility and saving time.

The system ensures **real-time updates**, organizes news by categories, and offers a user-friendly interface for seamless navigation. With features like **personalized recommendations** and **multi-source aggregation**, it provides a **comprehensive view** of current events while minimizing information overload.

Although the system successfully addresses many challenges in **news collection and summarization**, future enhancements can focus on **improving factual accuracy**, **bias detection**, and **scalability**. Integrating **advanced AI models**, **real-time verification**, and **cross-platform support** will further elevate the system's capabilities.

In conclusion, the **News Aggregator System** is a **robust and scalable** solution that simplifies access to diverse news sources while offering accurate and timely summaries. It holds significant potential for **continuous improvement** and **future innovation** in delivering high-quality, personalized news experiences.

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